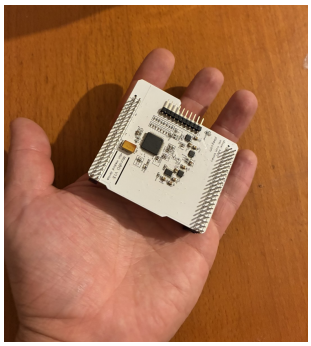
PCB & MANUFACTURING SPECIFICATION · SKU PIA24024P

# Board Material & Fabrication Data

Bare-board specification, mechanical features and the production quality test for MicroBCI (board revision V1.0). Circuit design files are proprietary and are not part of this document.



| FR-4 | 2 | 1.6 mm | 64.5 × 70 | 0.15 mm |
|---|---|---|---|---|
| Base material | Copper layers | Board thickness | mm outline | Minimum track |

## 01 BARE-BOARD FABRICATION PARAMETERS

Parameters for ordering the bare PCB, as used by common PCB order forms (Elecrow, JLCPCB, PCBWay).

| Parameter | Specification | Note |
|---|---|---|
| Base material | **FR-4**, standard Tg (130–140 °C) | Glass-epoxy laminate |
| Layers | **2** (top and bottom copper) | |
| Dimensions | **64.54 × 70.14 mm** | Contoured outline with notches and chamfers; routed profile |
| Thickness | **1.6 mm** | Standard |
| Solder mask colour | **White** | As on production boards |
| Silkscreen | **Black** | E1–E8, Ref, Biasout, pin numbers, warnings |
| Surface finish | **ENIG** recommended | Flat pads for the 0.5 mm-pitch TQFP-64; lead-free HASL acceptable |
| Copper weight | **1 oz** (35 µm) outer layers | |
| Minimum track / spacing | **0.15 mm (6 mil)** | Within standard capability |
| Minimum hole | Ø0.40 mm | Vias |
| RoHS | Lead-free materials and process | |

## 02 LAYER STACK-UP

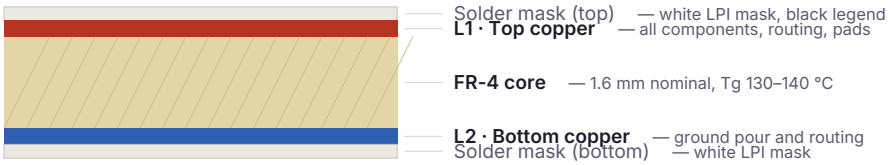



Figure 1 — 2-layer, 1.6 mm build. Layer roles are described in general terms only.


For research and education only. Not a medical device. Specifications subject to change.

## 03 MECHANICAL FEATURES

| Feature | Size | Qty | Purpose |
|---|---|---|---|
| Morpho socket holes | Ø1.02 mm plated | 76 | CN7 and CN10 sockets, 2 × 19 each, 2.54 mm |
| Electrode header holes | Ø1.00 mm plated | 10 | 1 × 10 electrode header, 2.54 mm |
| Test point holes | Ø0.70–0.80 mm plated | 3 | TP1–TP3 supply checks |
| Vias | Ø0.40 / Ø0.49 mm | 7 / 96 | Layer transitions |
| Mounting holes | — | 0 | Board held by the sockets |

## 04 ASSEMBLY

| | |
|---|---|
| **Technology** | Single-sided SMT (all parts on the top side) plus through-hole connectors |
| **Key IC** | TI ADS1299 (TQFP-64, 0.5 mm pitch) with low-noise on-board supplies |
| **Passives** | 0402 and 0805 chip parts; one tantalum bulk capacitor |
| **Connectors** | 2 × 2 × 19 sockets for the ST morpho headers; 1 × 10 electrode header |
| **Indicators** | 2 LEDs |
| **Soldering** | Lead-free reflow for SMD; selective or hand soldering for THT; no-clean flux, keep the analog input area free of residue |
| **Marking** | “MicroBCI V1.0”, “Power supply only from battery”, “Not medical device”, pieeg.com |

## 05 PRODUCTION QUALITY TEST

### 1 Supplies

Mount on a flashed NUCLEO-WB55 powered from a battery. **LED D4** must light; TP1, TP2, TP3 must read about −2.5 V, 3.3 V and +2.5 V.

### 2 Bluetooth

The board must advertise as EAREEG and stream 8 channels to the Python GUI or the Android app.

### 3 Shorted-input check

Short **Ref** to channels 1–8. Each channel must read close to 0 µV; open channels show noise.

## 06 HANDLING, STORAGE AND COMPLIANCE

| Topic | Requirement |
|---|---|
| ESD and storage | ESD-sensitive: handle by the edges on an anti-static surface. Store dry in the anti-static bag, 10–35 °C. |
| Environment | Indoor use only. No conformal coating; keep away from liquids and conductive gel. |
| Classification | Development and evaluation board for research and education. Not a medical device. |
| Design files | Schematic, layout, Gerbers and BOM are proprietary to PiEEG and are not published. |

## 07 RELATED DOCUMENTS

**MB-DS-001**
Product Datasheet

**MB-MD-002**
Mechanical & Dimensional Drawing

**MB-PIN-004**
Connector Pinout & Electrode Guide

**MB-QSG-005**
Quick Start & Safety Guide

**MB-SW-006**
Firmware, Bluetooth & Software Guide


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